

# Learning Style Classification

## Using Machine Learning in E-Learning Platforms

*Final Year Undergraduate Thesis · Dept. of CS & Engineering · East West University, Dhaka*

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### ABSTRACT

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Most e-learning platforms still rely on self-reported questionnaires to assess how students learn. The problem is straightforward: students fill them out once, often carelessly, and the label sticks regardless of how their actual behaviour evolves. This project sidesteps questionnaires entirely. Instead of asking students how they prefer to learn, it observes what they actually do on the platform and infers their learning style from that behaviour. The classification model is built around the Felder-Silverman Learning Style Model (FSLSM), which defines four independent dimensions: Active–Reflective, Sensing–Intuitive, Visual–Verbal, and Sequential–Global.

Two real-world datasets were used. CSMS (Computer Skills for Medical Students) provided 1,952 student records; CSHS (Computer Skills for Humanities Students) provided 670. Both capture clickstream behaviour — time spent on each resource, navigation paths, quiz patterns, and content-type preferences. These interaction traces were mapped onto the four FSLSM dimensions to construct the feature set. Before training, SMOTE was applied to handle class imbalance across dimensions, generating synthetic minority samples rather than simply discarding majority-class instances.

Four classifiers were built and compared: XGBoost, Decision Tree, K-Nearest Neighbours, and Support Vector Machine. Random Forest outperformed all of them. On CSMS it reached 94.33% accuracy; on CSHS, 93% — with consistent margins in precision, recall, and F1-score across both datasets. The advantage likely comes from how Random Forest handles non-linear feature interactions and tolerates residual class imbalance better than the other models tested. More importantly, the results confirm that clickstream data alone carries enough signal to classify learning styles reliably — no survey required.

The wider point is about what this makes possible. A platform that can infer learning style from behaviour can adjust content delivery accordingly — more visual material for visual learners, different sequencing for global vs. sequential types, adjusted pacing for active vs. reflective students. This project builds the classification layer that such a system needs as its foundation.

Several directions remain open. The dataset covers two academic disciplines; broader coverage across subjects and institutions would strengthen generalisation. Adding explainability tools — SHAP or LIME — would let educators see why a particular classification was made, which matters for trust and adoption. Sequential behaviour in learning sessions also points toward LSTM or Transformer architectures, which may capture temporal patterns that tree-based models miss. A practical end-point for this work is a lightweight LMS plugin or dashboard that makes real-time classification available to instructors without additional setup.

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## PROJECT SCOPE & CONTRIBUTIONS

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01

### Dataset Preparation & Feature Engineering

Behavioural features extracted from CSMS (1,952 instances) and CSHS (670 instances) clickstream logs; features aligned to all four FLSM dimensions. SMOTE applied to correct class imbalance before model training.

02

### Multi-Model Classification Study

Four classifiers benchmarked — XGBoost, Decision Tree, KNN, and SVM — with Random Forest achieving 94.33% (CSMS) and 93% (CSHS) accuracy, outperforming all baselines on precision, recall, and F1-score.

03

### FLSM Dimension Analysis

Independent classification pipeline built for each of the four FLSM dimensions (Active–Reflective, Sensing–Intuitive, Visual–Verbal, Sequential–Global), with per-dimension performance reported across all model types.

04

### Future Integration Roadmap

Proposed framework for real-time LMS integration, explainable AI layer using SHAP or LIME, and deep learning extensions (LSTM / Transformer) for sequential behaviour modelling in dynamic learning environments.

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**Keywords** — Machine Learning · Learning Style Classification · FLSM · Random Forest · SMOTE · E-Learning · Clickstream Analysis · Personalised Education · XAI · Adaptive Systems